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Where Minds Pollinate

Computer Science



Class 10

FBISE

Chapter 7

Exercise Solution

Follow Our Socials



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Computer Science 10th

Chapter no 7

Exercise (Q/A)

Short Q/A

Q2.1: Differentiate between basic literacy and digital literacy.

basic literacy	digital literacy
It is the term used for the ability to read, write and communicate information effectively.	It refers to the ability to effectively use digital technologies to access information.
It relies primarily on physical materials like paper textbooks, notebooks, and printed documents	It operates entirely within digital environments using smartphones, computers, and internet browsers.
It involves decoding vocabulary, understanding grammar, and writing by hand.	It requires technical skills like operating software, evaluating online sources for bias, and managing online data.
It is fundamental for learning and daily communication.	It is essential for success in the digital age (education, jobs, communication)

Q2.2: Suggest strategies for integrating digital literacy into computer graphics.

Strategies for integrating digital literacy into computer graphics:

- (i) **Teach Design Tools** – Train students to use tools like Canva or Photoshop.
- (ii) **Encourage Online Research** – Guide them to find and evaluate design ideas online.
- (iii) **Assign Projects** – Use real-world tasks like posters or social media graphics.
- (iv) **Use Collaborative Platforms** – Promote teamwork using tools like Figma or Google Slides.
- (v) **Teach Digital Ethics** – Explain copyright rules and responsible image use.
- (vi) **Create Digital Portfolios** – Let students showcase their work online.

These strategies help learners not only create strong visual content but also become confident and

responsible digital citizens.

Q2.3: Describe the role of books in verifying information for academic purposes.

Despite all the modern advancements in science and technology, books are still thought to be the most reliable and authentic source of information. Even in some fields of life, books are considered to be more reliable as compared to the internet and other web sources e.g., religion and history etc.

For example, you have an assignment to prepare a presentation on the topic of "اردو ادب میں خواتین کا کردار" (Role of Women in Urdu Literature). To prepare the presentation you will find the Urdu literature books from your school library. These books can be written about the women or by the women, who had prominent role in Urdu Literature. After collecting relevant data, you will organize them in the form of digital presentation by using some tools like MS-PowerPoint, Canva or any other software of your choice.

Q2.4: Discuss the relationship between communication, collaboration, and digital literacy.

Communication, collaboration, and digital literacy are deeply interconnected in the modern digital environment. Digital literacy refers to the ability to effectively use digital tools and technologies to find, evaluate, create, and share information. This skill set is essential for effective communication, as it enables people to use digital platforms such as emails, instant messaging, video conferencing, and social media to exchange ideas clearly and efficiently. Collaboration depends heavily on communication, especially in virtual or remote settings. With strong digital literacy skills, individuals can work together seamlessly using online tools like shared documents, project management software, and communication apps. Digital literacy ensures that participants know how to use these tools responsibly, understand digital etiquette, and manage information securely.

Q2.5: Do you think that X (Twitter) plays vital role in opinion making about any social issue? Support your answer with three arguments.

Yes, X (Twitter) plays a vital role in opinion making about social issues. It allows people to express views, spread awareness, and influence others quickly. Here are three arguments to support this:

Widespread Reach: X connects millions of users worldwide, making it a powerful platform to share opinions and bring attention to social issues instantly.

Hashtag Campaigns: Hashtags like #BlackLivesMatter have united voices and sparked global conversations, showing how X shapes public opinion.

Engagement with Influencers: Politicians, activists, and celebrities use X to share their stance on

issues, which influences how people think and respond. Thus, X is a major platform for raising awareness and shaping opinions on important social matters.

Q2.6: Outline effective strategies for identifying and countering online disinformation.

Identifying and countering online disinformation is essential for staying informed and digitally responsible. Here are some effective strategies:

Verify the Source: Always check if the information comes from a reliable and well-known source. Look for author credentials and official websites.

Cross-Check Information: Compare the content with trusted news outlets or official statements to confirm its accuracy.

Check the Date: Ensure the information is current and not outdated or taken out of context.

Look for Bias or Emotion: Be cautious of posts that use strong emotional language or appear one-sided, as they may be trying to mislead.

Use Fact-Checking Websites: Refer to platforms like Snopes, FactCheck.org, or AFP Fact Check to verify suspicious claims.

Report False Content: If you come across disinformation, report it to the platform so action can be taken against misleading content.

These strategies help promote safe and responsible use of digital media.

Q2.7: Differentiate between poster designing and billboard designing.

Poster designing	Billboard designing
Poster Designing is usually done on a smaller scale and is meant for close-up viewing.	Billboard Designing is done on a much larger scale and is meant to be seen from a distance, often by people in motion, such as drivers.
Posters are often used indoors for educational, promotional, or informational purposes, such as in classrooms, events, or public boards.	Billboards are placed outdoors in high-traffic areas and are designed to grab quick attention with bold visuals and minimal text.
Uses detailed combinations of shapes, text, and graphics.	Focuses on a highly highlighted key message with a company logo.

Q2.8: Define social media and describe the significance of YouTube and Facebook.

Social Media:

Social media refers to online platforms and websites that allow people to create, share, and interact with content and communicate with others. It enables users to connect globally, share ideas, opinions, videos, and images, and stay updated on various topics.

Significance of YouTube:

YouTube is a major platform for sharing video content. It is widely used for educational tutorials, skill development, entertainment, and spreading awareness. It allows users to learn new things visually and interact through comments and subscriptions.

Significance of Facebook:

Facebook is a social networking site used to connect with friends, share updates, photos, and videos. It plays a key role in promoting events, businesses, and community activities. It also supports group discussions, educational pages, and news sharing, making it a useful tool for both personal and public engagement.

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Exercise (Q/A)

Long Q/A

Q3.1: Explain the role of information collection in the field of digital literacy.

Role of Information Collection:

To communicate and publish key ideas and details to a variety of audiences, first we need to collect information to be communicated. This information can be collected through various sources. In other words, we must have some information in hand to communicate it. The role of digital literacy is how beautifully and effectively we can communicate this information to our target audience. Here we will discuss some common sources of information as follows:

Books:

Despite all the modern advancements in science and technology, books are still thought to be the most reliable and authentic source of information. Even in some fields of life, books are considered to be more reliable as compared to the internet and other web sources e.g. religion and history etc. Whenever you are supposed to communicate with a huge audience, create any digital content or find some information, you can collect information from books related to that topic. After collecting relevant data, you will organize them in the form of digital presentation by using some tools like MS-PowerPoint, Canva or any other software of your choice.

Web Sources:

The internet is the most common source of information now-a-days. Suppose you want to find some entertainment point for a person of any age or want to find or confirm some information, the internet is the most commonly available source. Let's suppose you want to find information on some latest international scholarships for the subject of computer science. To find this information type the search string "International undergraduate scholarships for Pakistani Students" on the Google home page. Explore the suggested results and open the web pages displayed as a search result.

Surveys and Google Forms:

All the required information is not always present in books or available online. Some information is supposed to be collected by you. For example, if you want to know about the popularity or success of a new course offered in your school, you cannot get information from the internet. The right people to provide you with the desired information are the students or teachers who are directly learning or teaching that course in your school. You can conduct a survey to collect that information. There are various methods to conduct a survey e.g.:

- Design a google form and distribute it online to collect information.
- Design a form in some document editor and distribute its printout to collect the information.

There are some key differences between survey and Google form.

Survey is used for:	Google form is used for:
Research purposes	Data collection on small scale
Data collection on a large scale	Event registration
Statistical analysis	Feedback collection
Data analysis	Quick polls and quizzes

Q3.2: Analyze the key components of digital literacy and their significance.

Key components of Digital Literacy:

There are certain components which play an important role in digital literacy. Some of them are as follows:

Digital Search / Information Literacy: It is the ability to collect, understand and evaluate information from various sources. Digital literacy begins with knowing how to find information online. You can find information by using search engines like Google, Bing, yahoo etc. or websites to look for relevant information.

- **Example:** If you want to find out how to solve a math problem, you can use Google to search for tutorials or watch a YouTube video explaining the steps.

Technical Skills: The technical knowledge and skill to use digital devices is required to communicate the information in digital form. The ability to use smart phone, digital cameras, computers, microphone and related devices is needed for digital literacy. Technical skills can be listed as follows:

- The ability to use computer and smart phones.
- The skill to navigate on internet and find relevant information while keeping your personal information and devices safe from cyber attacks.
- The ability to find and install relevant software and mobile applications to perform specific tasks, e.g. to install zoom, WhatsApp, or MS Teams to communicate with personal or official contacts to share digital information.
- Advanced skills include use of word processing and presentation software to develop digital content. To create vlogs, web pages, digital images and videos for social media marketing or interaction to communicate digital information.

Communication: The objective of digital literacy is to communicate the collected information to the target audience effectively. Good communication skills clearly express thoughts and opinions. It also includes appropriate online behavior, etiquette, netiquette and other digital communication protocols; Good digital communication skills include:

- Preparation of digital contents like images, videos
 - Preparation of word processing documents
 - Designing good PowerPoint presentations
 - Sending and receiving emails
 - Creating google forms, and google sheets for data collection
- All these digital content helps to communicate digital information by using digital devices like computer, smart phones etc.

Collaboration: The use of digital devices involves networking and collaboration with other people across distances. Online meetings, documents sharing, shared projects and activities are part of digital literacy. To develop digital citizenship and responsibility for correct information is also required in a healthy collaborative environment. Collaboration includes:

- Sending messages by using digital devices e.g. WhatsApp, Zoom, Google Chat etc.
- Editing and sharing google docs, sheets, slides and other real time collaboration platforms like LinkedIn, GitHub, Facebook, YouTube etc.

Digital Content Creation: The final product of the entire process is the digital content created for the audience. This content can be in the form of a graphical poster, Facebook post, YouTube video or some research paper. The other form of digital contents can be presentations, videos, interactive stories, online tutorials, webinars, audio books, blog posts, infographics, animations, quizzes, simulations, and games etc.

Ethics and Responsibility: The digital content has no boundaries. Therefore, higher standards of ethics and responsibility are required. These ethical standards involve cultural, social, and religious boundaries. Some important aspects of ethics are as follows:

- Ethical standards must respect cultural differences.
- Ensure that traditions, customs, and practices from different cultures are acknowledged and not misrepresented or disrespected.
- Content should be shared responsibly, ensuring it does not harm or offend different communities.
- Avoid disrespecting or infringing on the values of diverse faiths.

Overall, creators and users of digital content should be mindful of the global audience and act with responsibility and respect.

Q3.3: Define information extraction and its purpose.

Information extraction:

Information extraction is the process of automatically identifying and retrieving specific, meaningful data from large volumes of unstructured or semi-structured digital content such as web pages, emails, news articles, reports, and social media posts. Unlike structured data (such as data in tables or databases), unstructured data is more complex and harder to analyze directly, which is why information extraction is important.

Purpose:

- **Organizing Unstructured Data:** Most digital information is unstructured. Information extraction helps organize it into a structured form (like lists, databases, or spreadsheets) so that it can be more easily understood and analyzed.
- **Time Efficiency:** Instead of reading through long documents or webpages manually, information extraction allows users or systems to automatically pull out the most important facts – such as names, locations, dates, events, or keywords – saving time and effort.
- **Data Analysis and Decision-Making:** Extracted information can be analyzed for trends, patterns, or insights that are useful in research, business planning, policy-making, and more.
- **Support for Search Engines and AI:** Search engines use information extraction to provide relevant results. Similarly, digital assistants (like Siri or ChatGPT) rely on IE (Information Extraction) techniques to understand and respond accurately to user queries.
- **Academic and Professional Use:** In education, students and researchers use information extraction to summarize texts, cite key points, or gather information for reports and presentations. In professional settings, businesses use IE (Information Extraction) for market research, customer feedback analysis, and competitor monitoring.

Example: Suppose you are analyzing hundreds of online reviews about a mobile phone. Information extraction tools can automatically identify the phone's brand, key features mentioned (like battery life or camera), user sentiments (positive or negative), and common issues. This organized data helps manufacturers improve their products and researchers study consumer behavior.

Q3.4: Illustrate the use of images and graphics in billboard design.

Images and graphics play a vital role in billboard design by quickly grabbing attention and delivering a clear message to viewers passing by. Since billboards are usually placed in high-traffic areas like roadsides and highways, they must convey their message within seconds

- **Key Uses:**
 - (i) **Attracting Attention:** Bold, high-quality images or illustrations help catch the viewer's eye from a distance, even in a crowded visual environment.
 - (ii) **Supporting the Message:** Graphics can reinforce the written message. For example, an image of a cold drink on a hot day instantly communicates refreshment.
 - (iii) **Enhancing Visual Appeal:** Well-designed visuals make the billboard more attractive and memorable, increasing the chances that viewers will recall the product, service, or cause.
 - (iv) **Simplifying Communication:** Since there's limited space and viewing time, images help replace long text with visual storytelling, making the content easier to understand at a glance.
- **Example:** A billboard advertising a mobile phone may feature a large image of the phone with its screen glowing, paired with a short phrase like "Power That Lasts All Day." The image and text together make a strong and immediate impact.

Q3.5: Discuss the impact of social media on contemporary communication, referencing YouTube and WhatsApp as examples.

Social media has greatly transformed the way people communicate in the modern world. Platforms like YouTube and WhatsApp have made communication faster, more interactive, and global.

- **YouTube:**

YouTube allows users to share videos across the world. It enables visual communication on a massive scale, breaking language and cultural barriers through content such as tutorials, vlogs, news, and entertainment. It also supports real-time feedback through comments and live streams, fostering two-way communication between content creators and viewers

- **WhatsApp:**

WhatsApp has revolutionized personal and group communication by offering instant messaging, voice calls, video calls, and file sharing. It allows real-time interaction regardless of distance and supports end-to-end encryption, ensuring privacy. Its group chat feature also makes teamwork, family communication, and class communication easier and more effective.

Conclusion:

Social media platforms like YouTube and WhatsApp have made communication faster, more visual, interactive, and accessible globally. They bridge geographical gaps and allow information sharing in ways that were impossible in the past, greatly impacting personal, educational, and professional communication.